# A First Course in Abstract Algebra

## Section 29: Introduction to Extension Fields

Detailed worked solutions

Mathjock

**Text and numbering.** John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

**How to read the solutions.** Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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# 29 Introduction to Extension Fields

### Exercise 29.1: A shifted square root is algebraic

For $\alpha = 1 + \sqrt{2}$, we have $(\alpha - 1)^2 = 2$. Expanding gives $\alpha^2 - 2\alpha - 1 = 0$. Thus the nonzero rational polynomial $f(x) = x^2 - 2x - 1$ witnesses algebraicity over $\mathbb{Q}$.

### Exercise 29.2: A sum of two square roots

For $\alpha = \sqrt{2} + \sqrt{3}$, square to obtain $\alpha^2 = 5 + 2\sqrt{6}$. Isolate the radical and square again:

$$(\alpha^2 - 5)^2 = 24.$$

Hence $\alpha^4 - 10\alpha^2 + 1 = 0$, so $f(x) = x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$ proves algebraicity. For this exercise an annihilating polynomial suffices; minimality is not required.

### Exercise 29.3: A shifted imaginary unit

For $\alpha = 1 + i$, the equation $(\alpha - 1)^2 = -1$ gives $\alpha^2 - 2\alpha + 2 = 0$. Therefore $x^2 - 2x + 2$ is a nonzero rational polynomial vanishing at $\alpha$.

### Exercise 29.4: A nested square and cube root

Let $\alpha = \sqrt{1 + \sqrt[3]{2}}$. Then $\alpha^2 - 1 = \sqrt[3]{2}$, so $(\alpha^2 - 1)^3 = 2$. Thus

$$f(x) = (x^2 - 1)^3 - 2 = x^6 - 3x^4 + 3x^2 - 3$$

is a rational annihilating polynomial. Every displayed operation is applied to the actual given number, so no converse claim about extraneous roots is needed.

### Exercise 29.5: A complex nested radical

For $\alpha = \sqrt{\sqrt[3]{2} - i}$, we have $(\alpha^2 + i)^3 = 2$. Set $A(x) = (x^2 + i)^3 - 2$. Multiplying by its coefficient-conjugate gives a polynomial in $\mathbb{Q}[x]$:

$$A(x)\overline{A}(x) = x^{12} + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5.$$

The first factor vanishes at $\alpha$, so the product does too. For checking the multiplication, write $A(x) = (x^6 - 3x^2 - 2) + i(3x^4 - 1)$; the product is the sum of squares of these two real polynomials.

### Exercise 29.6: Minimal polynomial of a nested quadratic radical

Let $\alpha = \sqrt{3 - \sqrt{6}}$. Then $(\alpha^2 - 3)^2 = 6$, giving $p(x) = x^4 - 6x^2 + 3$. To prove minimality, first $\sqrt{6} = 3 - \alpha^2 \in \mathbb{Q}(\alpha)$. In $K = \mathbb{Q}(\sqrt{6})$, the element $3 - \sqrt{6}$ is not a square: the norm of a square $(u + v\sqrt{6})^2$ is $(u^2 - 6v^2)^2$, a rational square, but the norm of $3 - \sqrt{6}$ is $9 - 6 = 3$, not a rational square. Thus $\alpha \notin K$ and $[K(\alpha) : K] = 2$. Since $[K : \mathbb{Q}] = 2$, $\alpha$ has degree four. The monic annihilating quartic must therefore be $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$.

### Exercise 29.7: A rational shift inside a nested radical

For $\alpha = \sqrt{1/3 + \sqrt{7}}$, the equation $(\alpha^2 - 1/3)^2 = 7$ gives

$$p(x) = x^4 - \frac{2}{3}x^2 - \frac{62}{9}.$$

The field $\mathbb{Q}(\alpha)$ contains $K = \mathbb{Q}(\sqrt{7})$ because $\sqrt{7} = \alpha^2 - 1/3$. The norm of $1/3 + \sqrt{7}$ down to $\mathbb{Q}$ is $1/9 - 7 = -62/9$. It cannot be the norm of a square in $K$, since such a norm is a square of a rational number and hence nonnegative. Therefore adjoining $\alpha$ to $K$ has degree two, and the full degree is four. Thus $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$ and $\deg(\alpha, \mathbb{Q}) = 4$. Multiplying by nine gives an integer polynomial but the irreducible polynomial is conventionally monic.

### Exercise 29.8: Minimal polynomial of a real-imaginary sum

Let $\alpha = \sqrt{2} + i$. Its equation $(\alpha^2 - 1)^2 = -8$ gives $p(x) = x^4 - 2x^2 + 9$. For minimality, $3/\alpha = \sqrt{2} - i$ because $(\sqrt{2} + i)(\sqrt{2} - i) = 3$. Hence

$$\sqrt{2} = \frac{\alpha + 3/\alpha}{2}, \qquad i = \frac{\alpha - 3/\alpha}{2}.$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, i)$. The real field $\mathbb{Q}(\sqrt{2})$ does not contain $i$, so adjoining $i$ doubles its degree two, giving degree four. Therefore $p$ is the monic minimal polynomial and $\deg(\alpha, \mathbb{Q}) = 4$.

### Exercise 29.9: The imaginary unit over the rationals

The number $i$ satisfies $x^2 + 1 = 0$. This quadratic has no rational root (indeed no real root), so it is irreducible over $\mathbb{Q}$. Thus $i$ is algebraic of degree two over $\mathbb{Q}$.

### Exercise 29.10: A nonreal number over the reals

The number $1 + i$ satisfies $x^2 - 2x + 2 = 0$. Its discriminant is $-4 < 0$, so the polynomial is irreducible over $\mathbb{R}$. Thus $1 + i$ is algebraic of degree two over $\mathbb{R}$.

### Exercise 29.11: A square root of a transcendental number

The number $\sqrt{\pi}$ is transcendental over $\mathbb{Q}$. If it were algebraic, its square would be algebraic because algebraic elements form a field (or because powers of an algebraic element lie in its finite simple extension). This would make $\pi$ algebraic over $\mathbb{Q}$, contradicting the known transcendence of $\pi$.

### Exercise 29.12: The same number over the reals

Since $\sqrt{\pi} \in \mathbb{R}$, the polynomial $x - \sqrt{\pi} \in \mathbb{R}[x]$ vanishes at it. Its minimal polynomial over $\mathbb{R}$ is linear and its degree is one. Algebraicity always depends on the base field.

### Exercise 29.13: A square root over a rational-function field

Put $t = \sqrt{\pi}$. Since $\pi = t^2$, the base field is $\mathbb{Q}(t^2)$. The polynomial $x^2 - \pi$ shows degree at most two. It has no zero in $\mathbb{Q}(\pi)$: if a rational function $P(\pi)/Q(\pi)$ squared to $\pi$, transcendence would give the polynomial identity $P(T)^2 = TQ(T)^2$. The exponent of $T$ on the left is even, and on the right odd, a contradiction after removing any common powers. Thus the degree is two.

### Exercise 29.14: A square of pi over the rationals

The number $\pi^2$ is transcendental over $\mathbb{Q}$. If a nonzero $f \in \mathbb{Q}[x]$ satisfied $f(\pi^2) = 0$, the nonzero polynomial $f(x^2)$ would vanish at $\pi$, contradicting transcendence. Therefore no algebraic degree is assigned.

### Exercise 29.15: An element already in the base field

Here $\pi^2 \in \mathbb{Q}(\pi)$, so its minimal polynomial over that field is $x - \pi^2$ and its degree is one. The coefficient $-\pi^2$ is allowed because it lies in the specified base field.

### Exercise 29.16: Two different powers of a transcendental number

Set $T = \pi^3$. The element $\pi^2$ satisfies $x^3 - T^2 = 0$ over $\mathbb{Q}(T)$. A reducible cubic over a field has a root in that field. But a rational root $P(T)/Q(T)$ would imply $P(T)^3 = T^2 Q(T)^3$. Comparing exponents of the irreducible factor $T$ gives a multiple of three on the left and an integer congruent to two modulo three on the right. This is impossible. Hence the cubic is irreducible, and $\pi^2$ has degree three over $\mathbb{Q}(\pi^3)$.

### Exercise 29.17: Splitting the four-element-field quadratic

In characteristic two, $-\alpha = \alpha$ and $\alpha^2 = \alpha + 1$. Polynomial division gives

$$x^2 + x + 1 = (x + \alpha)(x + \alpha + 1),$$

since the sum of the roots is 1 and their product is $\alpha(\alpha + 1) = \alpha^2 + \alpha = 1$. Thus the second root is $\alpha + 1$.

### Exercise 29.18: Constructing the nine-element field

**(a)** The values of $x^2 + 1$ at $0, 1, 2$ in $\mathbb{Z}_3$ are $1, 2, 2$, so it has no root and is irreducible. **(b)** With $\alpha^2 = -1 = 2$, each element is uniquely $a + b\alpha$ for $a, b \in \mathbb{Z}_3$. Use

$$(a + b\alpha) + (c + d\alpha) = (a + c) + (b + d)\alpha,$$

$$(a + b\alpha)(c + d\alpha) = (ac - bd) + (ad + bc)\alpha,$$

reducing coefficients modulo three. For example $(1+2\alpha)(2+\alpha) = 2+5\alpha+2\alpha^2 = 2\alpha$. In the requested order, the full tables are

| $+$ | $0$ | $1$ | $2$ | $\alpha$ | $2\alpha$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $2+2\alpha$ |
|---|---|---|---|---|---|---|---|---|---|
| $0$ | $0$ | $1$ | $2$ | $\alpha$ | $2\alpha$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $2+2\alpha$ |
| $1$ | $1$ | $2$ | $0$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $2+2\alpha$ | $\alpha$ | $2\alpha$ |
| $2$ | $2$ | $0$ | $1$ | $2+\alpha$ | $2+2\alpha$ | $\alpha$ | $2\alpha$ | $1+\alpha$ | $1+2\alpha$ |
| $\alpha$ | $\alpha$ | $1+\alpha$ | $2+\alpha$ | $2\alpha$ | $0$ | $1+2\alpha$ | $1$ | $2+2\alpha$ | $2$ |
| $2\alpha$ | $2\alpha$ | $1+2\alpha$ | $2+2\alpha$ | $0$ | $\alpha$ | $1$ | $1+\alpha$ | $2$ | $2+\alpha$ |
| $1+\alpha$ | $1+\alpha$ | $2+\alpha$ | $\alpha$ | $1+2\alpha$ | $1$ | $2+2\alpha$ | $2$ | $2\alpha$ | $0$ |
| $1+2\alpha$ | $1+2\alpha$ | $2+2\alpha$ | $2\alpha$ | $1$ | $1+\alpha$ | $2$ | $2+\alpha$ | $0$ | $\alpha$ |
| $2+\alpha$ | $2+\alpha$ | $\alpha$ | $1+\alpha$ | $2+2\alpha$ | $2$ | $2\alpha$ | $0$ | $1+2\alpha$ | $1$ |
| $2+2\alpha$ | $2+2\alpha$ | $2\alpha$ | $1+2\alpha$ | $2$ | $2+\alpha$ | $0$ | $\alpha$ | $1$ | $1+\alpha$ |

| $\cdot$ | $0$ | $1$ | $2$ | $\alpha$ | $2\alpha$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $2+2\alpha$ |
|---|---|---|---|---|---|---|---|---|---|
| $0$ | $0$ | $0$ | $0$ | $0$ | $0$ | $0$ | $0$ | $0$ | $0$ |
| $1$ | $0$ | $1$ | $2$ | $\alpha$ | $2\alpha$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $2+2\alpha$ |
| $2$ | $0$ | $2$ | $1$ | $2\alpha$ | $\alpha$ | $2+2\alpha$ | $2+\alpha$ | $1+2\alpha$ | $1+\alpha$ |
| $\alpha$ | $0$ | $\alpha$ | $2\alpha$ | $2$ | $1$ | $2+\alpha$ | $1+\alpha$ | $2+2\alpha$ | $1+2\alpha$ |
| $2\alpha$ | $0$ | $2\alpha$ | $\alpha$ | $1$ | $2$ | $1+2\alpha$ | $2+2\alpha$ | $1+\alpha$ | $2+\alpha$ |
| $1+\alpha$ | $0$ | $1+\alpha$ | $2+2\alpha$ | $2+\alpha$ | $1+2\alpha$ | $2\alpha$ | $2$ | $1$ | $\alpha$ |
| $1+2\alpha$ | $0$ | $1+2\alpha$ | $2+\alpha$ | $1+\alpha$ | $2+2\alpha$ | $2$ | $\alpha$ | $2\alpha$ | $1$ |
| $2+\alpha$ | $0$ | $2+\alpha$ | $1+2\alpha$ | $2+2\alpha$ | $1+\alpha$ | $1$ | $2\alpha$ | $\alpha$ | $2$ |
| $2+2\alpha$ | $0$ | $2+2\alpha$ | $1+\alpha$ | $1+2\alpha$ | $2+\alpha$ | $\alpha$ | $1$ | $2$ | $2\alpha$ |

Irreducibility ensures this quotient is a field, so each nonzero row of the multiplication table contains 1.

### Exercise 29.19: Correcting algebraic element

An element $\alpha$ of an extension field $E/F$ is algebraic over $F$ exactly when it is a zero of a *nonzero polynomial in* $F[x]$. Both qualifications matter: the zero polynomial vanishes everywhere, and a polynomial with arbitrary coefficients in $E$ could always be $x - \alpha$.

### Exercise 29.20: Correcting transcendental element

An element $\beta \in E$ is transcendental over $F$ exactly when no *nonzero* polynomial in $F[x]$ vanishes at $\beta$. The zero polynomial is excluded, since it vanishes at every element.

### Exercise 29.21: Correcting monic polynomial

A nonzero polynomial is monic when its *leading coefficient* is 1; its other coefficients may be arbitrary. For example $x^2 - 3x + 7$ is monic although not all its coefficients are one.

### Exercise 29.22: Correcting simple extension

The extension $E/F$ is simple if $E = F(\alpha)$ for some $\alpha \in E$. Equivalently no proper subfield of $E$ that contains *both* $F$ *and* $\alpha$ exists. Omitting the condition that the subfield contain $F$ changes the

meaning.

### Exercise 29.23: True and false statements

(a) True. The transcendence of $\pi$ over $\mathbb{Q}$ is a known theorem used here.

(b) True. $\mathbb{C} = \mathbb{R}(i)$.

(c) True. For $a \in F$, the nonzero polynomial $x - a \in F[x]$ vanishes at $a$.

(d) True. The rational field is a subfield of $\mathbb{R}$.

(e) False. A subfield has the same identity and characteristic; $\mathbb{Q}$ has characteristic zero and $\mathbb{Z}_2$ characteristic two.

(f) True. The minimal polynomial divides every rational annihilating polynomial, so their degrees are at least $n$.

(g) False. Take $\alpha = \sqrt{2}$, of degree two over $\mathbb{Q}$, and $f = x - \sqrt{2} \in \mathbb{R}[x]$, of degree one.

(h) True. Kronecker's construction adjoins a zero using a quotient by an irreducible factor.

(i) False. $x^2 + 1 \in \mathbb{R}[x]$ has no zero in the extension $\mathbb{R}$ itself.

(j) True. Evaluation $\mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is onto and is injective by transcendence of $\pi$.

### Exercise 29.24: Making transcendental numbers algebraic over larger fields

**(a)** Take $F = \mathbb{Q}(\pi^3)$. The polynomial $x^3 - \pi^3$ vanishes at $\pi$, and has no root in $F$: treating $T = \pi^3$ as an indeterminate, $P(T)^3 = TQ(T)^3$ contradicts exponents of $T$. A cubic with no root is irreducible. Hence the degree is three. **(b)** Take $E = \mathbb{Q}(e^{10})$. The element $u = e^2$ is transcendental over $\mathbb{Q}$, since otherwise $e$ would be algebraic over $\mathbb{Q}$. For any transcendental $u$, put $L = \mathbb{Q}(u^m)$ and let $A$ be the $L$-span of $1, u, \ldots, u^{m-1}$. These powers are independent: after clearing denominators in a proposed relation, powers occurring in different terms have different exponent residues modulo $m$, so no nonzero polynomial relation can vanish at the transcendental $u$. Thus $\dim_L A = m$. Reducing powers using $u^m \in L$ shows that $A$ is closed under multiplication and contains 1. For $0 \neq a \in A$, multiplication by $a$ is an injective linear map $A \to A$, because $A$ lies in the field $\mathbb{Q}(u)$. Finite dimension makes it surjective, so $ab = 1$ for some $b \in A$. Hence $A$ is a field. It contains $L$ and $u$ and is contained in $\mathbb{Q}(u)$, so $A = \mathbb{Q}(u)$ and $[\mathbb{Q}(u) : \mathbb{Q}(u^m)] = m$. Taking $m = 5$ proves $e^2$ has degree five over $E$.

### Exercise 29.25: Splitting an irreducible binary cubic

**(a)** The cubic $p = x^3 + x^2 + 1$ takes value 1 at both elements $0, 1$ of $\mathbb{Z}_2$, so has no linear factor and is irreducible. **(b)** If $p(\alpha) = 0$, then $\alpha^3 = \alpha^2 + 1$. Squaring an equation in characteristic two shows that $\alpha^2$ and $\alpha^4$ are roots too. They are distinct: $\alpha$ has multiplicative order seven in the field of eight elements, so its first, second, and fourth powers differ. Reducing powers gives $\alpha^4 = \alpha^2 + \alpha + 1$. Thus

$$p(x) = (x + \alpha)(x + \alpha^2)(x + \alpha^2 + \alpha + 1).$$

The first long division gives $p(x) = (x + \alpha)(x^2 + (1 + \alpha)x + \alpha^2 + \alpha)$; the other two displayed roots factor that quadratic. This supplies the actual linear factors, not just their existence.

### Exercise 29.26: The additive and multiplicative groups of eight elements

Degree three over $\mathbb{Z}_2$ gives the unique representation $a + b\alpha + c\alpha^2$ with binary coefficients. Addition is coordinatewise, so the additive group is $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. There are seven nonzero elements. Their multiplicative group has prime order seven, so any nonidentity element generates it by Lagrange's theorem. Hence it is $\mathbb{Z}_7$. The additive group is not cyclic: every nonzero element has additive order two.

### Exercise 29.27: Why minimal polynomial is an appropriate name

Division by the monic irreducible polynomial $p = \operatorname{irr}(\alpha, F)$ shows that any $f \in F[x]$ with $f(\alpha) = 0$ is a multiple of $p$. Indeed its remainder would have smaller degree and still vanish at $\alpha$, so must be zero. Thus $p$ has the smallest possible degree among nonzero annihilating polynomials, and monicity makes it unique.

### Exercise 29.28: Synopsis of Kronecker construction

Choose an irreducible factor $p$ of the given nonconstant $f \in F[x]$ and form the field $E = F[x]/(p)$. The constant cosets embed $F$ in $E$, and the coset $\alpha = x + (p)$ satisfies $p(\alpha) = 0$, hence $f(\alpha) = 0$ because $p$ divides $f$.

### Exercise 29.29: Exchanging a transcendental generator

Since $\alpha$ is algebraic over $F(\beta)$, clear denominators in an annihilating equation to obtain

$$H(\alpha, \beta) = 0, \qquad H(X, Y) \in F[X, Y] \setminus \{0\}.$$

Here clearing denominators can be done without making every coefficient zero when evaluated at $\beta$; start with actual nonzero coefficients in $F(\beta)$ and choose polynomial representatives. Regard $H(\alpha, Y)$ as a polynomial over $F(\alpha)$. It is not the zero polynomial: some coefficient is a nonzero polynomial in $\alpha$, and transcendence of $\alpha$ prevents its vanishing. Its value at $Y = \beta$ is zero. Its degree in $Y$ cannot be zero either, because that would give a nonzero element equal to zero. Thus $\beta$ is algebraic over $F(\alpha)$.

### Exercise 29.30: Counting a finite simple extension

If $\operatorname{irr}(\alpha, F)$ has degree $n$, every element of $F(\alpha)$ has a unique expression

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}, \qquad a_i \in F.$$

Spanning follows from division by the minimal polynomial, and uniqueness from its minimal degree. Each of the $n$ coefficients has $q$ choices, independently. Thus there are exactly $q^n$ elements.

### Exercise 29.31: A twenty-seven-element field

**(a)** The polynomial $x^3 + 2x + 1$ takes value 1 at each of $0, 1, 2$ in $\mathbb{Z}_3$, so has no root and is irreducible.
**(b)** Therefore $\mathbb{Z}_3[x]/(x^3 + 2x + 1)$ is a field. Each coset has a unique representative $a + bx + cx^2$ with $a, b, c \in \mathbb{Z}_3$, giving $3^3 = 27$ elements.

### Exercise 29.32: Quadratic extensions in every prime characteristic

**(a)** For odd $p$, the nonzero residues pair as $\{a, -a\}$, with two distinct members per pair. Their squares agree, and equality of squares implies this same pairing because $(a-b)(a+b) = 0$ in the field. Thus there are only $(p-1)/2$ nonzero squares, fewer than $p-1$, so a nonsquare exists. **(b)** Choose a nonsquare $d \in \mathbb{Z}_p$. Then $x^2 - d$ has no root and is irreducible, and its quotient field has $p^2$ elements. For the omitted prime $p = 2$, use $x^2 + x + 1$, which has no root in $\mathbb{Z}_2$ and produces four elements. This covers every prime.

### Exercise 29.33: Nonconstant rational functions stay transcendental

Let $\beta = f(\alpha)/g(\alpha) \notin F$, where $f, g \in F[x]$ and $g(\alpha) \neq 0$. The polynomial $f(x) - \beta g(x) \in F(\beta)[x]$ vanishes at $\alpha$. It is nonzero, since otherwise any nonzero coefficient of $g$ would show $\beta \in F$. Thus $\alpha$ is algebraic over $F(\beta)$. If $\beta$ were algebraic over $F$, adjoining it gives a finite extension, and then adjoining $\alpha$ gives another finite extension. Their composite would make $\alpha$ algebraic over $F$, contradicting its transcendence. Hence $\beta$ is transcendental over $F$.

### Exercise 29.34: A cubic-radical field without checking every axiom

The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at 2. Therefore $\mathbb{Q}(\sqrt[3]{2})$ has basis $1, \sqrt[3]{2}, (\sqrt[3]{2})^2$ and consists exactly of the displayed expressions with rational coefficients. Since it is a simple algebraic extension, it is already a field. This identifies the set as a subfield of $\mathbb{R}$ without separately proving formulas for inverses.

### Exercise 29.35: Fields of orders eight, sixteen, and twenty-five

For order eight use $\mathbb{Z}_2[x]/(x^3 + x + 1)$; the cubic has no binary root, so is irreducible. For order sixteen use $\mathbb{Z}_2[x]/(x^4 + x + 1)$. It has no binary root. A reducible quartic with no linear factor would be a product of irreducible quadratics, but the only monic irreducible binary quadratic is $x^2 + x + 1$. Modulo that quadratic, $x^3 = 1$ and $x^4 = x$, so $x^4 + x + 1$ has remainder 1 and is not divisible by it. Hence it is irreducible. For order twenty-five use $\mathbb{Z}_5[x]/(x^2 + 2)$; a root would have square 3, which is not among $0, 1, 4$. Unique remainders of degrees less than $3, 4, 2$ give respectively $2^3, 2^4, 5^2$ elements.

### Exercise 29.36: Every element of a finite field is algebraic

Let $|F| = q$. Its nonzero elements form a group of order $q-1$, so Lagrange's theorem gives $a^{q-1} = 1$ for every $a \neq 0$. Thus all elements, including zero, satisfy $a^q - a = 0$. The nonzero polynomial $x^q - x$ has coefficients in the prime subfield $\mathbb{Z}_p$, so every element is algebraic over that subfield.

### Exercise 29.37: Prime-power order

List the finitely many elements as $a_1, \ldots, a_r$ and form the chain $F_0 = \mathbb{Z}_p$, $F_i = F_{i-1}(a_i)$. By Exercise 36, each $a_i$ is algebraic even over $\mathbb{Z}_p$, hence also over $F_{i-1}$. If its degree there is $d_i$, Exercise 30 gives $|F_i| = |F_{i-1}|^{d_i}$. Starting from $|F_0| = p$ and iterating yields $|F| = p^{d_1 \cdots d_r}$, a prime power. Elements already adjoined simply contribute $d_i = 1$.